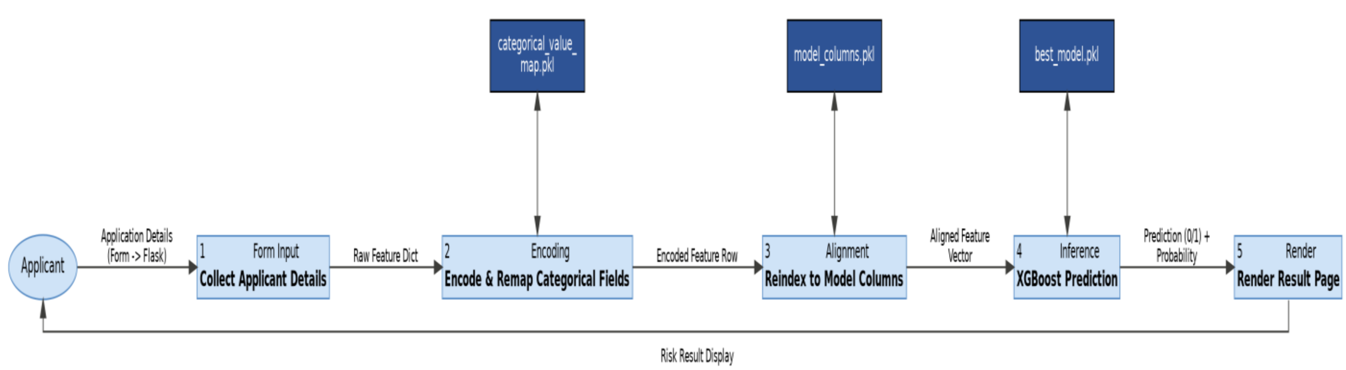
Data Flow Diagram

Date	15/07/2026
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	2 Marks

Data Flow Diagram (DFD) details and process descriptions:

Symbol	Name	Description	Example in Credit Card Approval Prediction System
Oval / Rounded shape	External Entity	Interacts with the system.	Applicant, Credit Analyst
Rectangle with numbered header	Process	Transforms input data.	Form Input, Feature Encoding, Column Alignment, XGBoost Inference, HTML Render
Rectangle (solid fill, no number)	Data Store	Stores persistent information.	best_model.pkl, model_columns.pkl, categorical_value_map.pkl, application_record.csv, credit_record.csv
Labeled arrow	Data Flow	Movement of data.	Application Details (Form -> Flask), Aligned Feature Vector (Flask -> Model)

Diagram



Data Flow Steps:

1. The Applicant enters details through the Web Form (predict.html). This forms the raw applicant data.

2. The data flows into the Flask Server (app.py), where rare occupation values are remapped to "Other" and categorical fields are encoded to match the training-time format - referencing categorical_value_map.pkl.

3. The encoded feature row is reindexed against model_columns.pkl, aligning it exactly to the column order the model was trained on (missing dummy columns filled with 0).

4. The aligned feature vector flows into the tuned XGBoost Model (best_model.pkl), which returns a prediction (0 = low-risk, 1 = high-risk) and a risk probability.

5. Flask translates the output into a prediction label and probability, and renders it on result.html for the Applicant.